QUASI-COHERENT SUBMODULES AND QUOTIENTS OF A COHERENT MODULE ON A LOCALLY NOETHERIAN ALGEBRAIC SPACE ARE COHERENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The pullback along an etale morphism is an exact restriction functor. Properties of Spaces, equation spaces-properties-equation-restrict-modules (invoked at proof lines 2162-2164). It makes pullback along a surjective etale $f : U \to X$ exact, so that the pullback of a submodule or a quotient is again a submodule or quotient.
- Coherence is detected after an etale cover. Same chapter spaces-cohomology.tex, label lemma-coherent-Noetherian (invoked at proof line 2166). It reduces the question whether a module on X is coherent to the corresponding question for its pullback to the scheme U.
- The statement for schemes. Cohomology of Schemes, label coherent-lemma-coherent-Noetherian-quasi-coherent-sub-quotient (invoked at proof line 2170). It supplies the result on the locally Noetherian scheme U, which then descends to X.

Source reference. Original lemma 01Y1: sources/stacks-chow/tags.txt, source lines 2540–2540. Navigation pointer for \ref{coherent-lemma-coherent-Noetherian-quasi-coherent-sub-quotient} used inside the counted span (line 2171); the argument that consumes it is retained in the span.

Source reference. Original lemma 07UB: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 2094–2094. Navigation pointer for \ref{lemma-coherent-Noetherian} used inside the counted span (line 2166); the argument that consumes it is retained in the span.

Date: stacks-project:spaces-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original equation 03LW: sources/stacks-chow/tags.txt, source lines 4728–4728. Navigation pointer for \ref{spaces-properties-equation-restrict-modules} used inside the counted span (line 2164); the argument that consumes it is retained in the span.

Lemma 0.1. *Let S be a scheme. Let X be a locally Noetherian algebraic space over S. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module. Any quasi-coherent submodule of $\mathcal{F}$ is coherent. Any quasi-coherent quotient module of $\mathcal{F}$ is coherent.*

Proof. Choose a scheme U and a surjective étale morphism $f : U \to X$. Pullback f^* is an exact functor as it equals a restriction functor, see Properties of Spaces, Equation (03LW). By Lemma 07UB we can check whether an $\mathcal{O}_X$-module $\mathcal{G}$ is coherent by checking whether $f^*\mathcal{H}$ is coherent. Hence the lemma follows from the case of schemes which is Cohomology of Schemes, Lemma 01Y1. □